

ResumeGenAI: Supporting Job Seekers with LLM-Driven Resume Feedback

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Abstract

Rising unemployment and rapid advances in AI have disrupted the job market, making it more important than ever for job seekers to present themselves effectively—starting with their resumes. While prior research has explored resume evaluation, little attention has been given to leveraging AI for delivering personalized, detailed, instant and actionable feedback. We present ResumeGenAI, an online platform designed to support job seekers create resumes, personalize them to specific job roles, compare them, and seek help through a conversational interface. Through a three-week study ($N = 34$), we evaluated its effectiveness in helping job seekers build resumes. We found that the platform significantly improved resume quality in areas such as skills, grammar, and overall impact, as assessed by expert reviewers. Participants also highlighted features like resume personalization, export analysis, and resume templates as particularly valuable.

CCS Concepts

• **Social and professional topics** → **Employment issues**; • **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → **Natural language processing**.

Keywords

Resume Feedback, Resume Screening, Large Language Model, Employment, Artificial Intelligence

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1 Introduction

In January 2025, the OECD reported an overall unemployment rate of 4.8% [25]. In comparison, the latest available figure for long-term unemployment measured in 2022 is 25.44 % [24]. A recent study [38]

reveals that 32.8% of the occupations could be fully affected by AI technologies such as ChatGPT, while 36.5% might experience a partial impact. In the face of unemployment and disruptive AI advancements, it becomes critical to how job seekers present themselves to recruiters. In that direction, a resume is the first thing that a recruiter looks at. According to an eye tracking study done in 2018 [15], it was concluded that a recruiter looks at the resume for an average of 7.4 seconds. In those 7.4 seconds, the recruiter decides whether to proceed with the applicant for the next round or not. Hence, crafting a resume that precisely highlights relevant skills and experience while maintaining clarity, language, and structure becomes a necessity.

Although peer and expert guidance can be valuable in crafting a good resume, it is often time consuming, inconsistent, and not immediately accessible. As a result, job seekers still struggle to iteratively enhance narrative flow, tailor their resumes to specific roles, or obtain nuanced advice on grammar, layout, and overall impact.

Many previous works [10, 13, 14, 18, 23, 31, 36] have provided solutions for resume evaluation. Recent advances in large language models (LLMs) have enabled deeper context-aware analysis [22]. However, most existing solutions focus on a narrow scope, producing scores or comments on skill extraction or job matching, rather than providing comprehensive resume feedback. For instance, CareerMapper [14] evaluates resume completeness by comparing a user's resume with similar LinkedIn profiles [16], highlighting commonly missing sections and providing recommendations to match resumes to job descriptions. CareerCounselling [31] initiates a chat session based on the user's resume where the user can ask resume improvement-related questions; however, users have to figure out the issues themselves and frame the questions, which limits the usefulness of the tool to their existing knowledge and awareness of resume best practices. Most of the existing systems offer generic evaluations lacking contextual, section-specific, and actionable feedback tailored to a candidate's background, target role, writing style, and content.

To address these gaps, we introduce ResumeGenAI, a platform that uses an LLM to provide detailed, actionable feedback along with guidance on how to improve a resume on various attributes. Beyond providing feedback, ResumeGenAI offers job-specific customization and resume comparisons across multiple versions. ResumeGenAI also employs a conversational user interface (CUI) that enables dialogue-based real-time interactions, allowing users to iteratively refine their resumes through natural back-and-forth queries. It

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provides an immediate and user-centric way for job seekers to explore ‘what if’ scenarios, clarify suggestions, etc. We assess how effectively ResumeGenAI can support job seekers’ need for building their resume as compared to the traditional feedback received from peers, or with the help of the Internet.

In this work, we make the following contributions:

- Design of an online resume improvement platform, ResumeGenAI, to support job seekers by providing detailed actionable feedback on various attributes considered essential in a resume.
- Empirical results of a three-week study with ResumeGenAI. Our results demonstrate that the platform significantly improved the quality of attributes like skills, grammar and overall impact in job seekers’ resumes.

2 ResumeGenAI App

ResumeGenAI’s design is aimed at supporting job seekers with resume building and enhancement. The goal was to leverage Generative AI’s capabilities to provide instant actionable feedback and provide an opportunity to personalize the feedback to job seekers’ specific needs. ResumeGenAI uses the following technologies/frameworks: (i) GPT-4o [27] as the foundational LLM; (ii) Django [7] as a web development framework; (iii) MongoDB [20] as a database; (iv) Celery [29] as asynchronous task scheduler.

2.1 ResumeGenAI Features

ResumeGenAI was designed as a web application accessible on any device for job seekers. Job seekers can upload their resumes to receive actionable feedback on various aspects, personalize their resumes for specific job roles or postings, and get instant answers to their questions through a conversational agent, among other features. We use LLM capabilities to thoroughly analyze a resume and provide insights and actionable feedback to improve. First, we use LLMs to parse the resume into a structured JSON format, and then we use this structured data to analyze and score the resume. This process is done using a set of iteratively refined prompts to obtain optimal results. The prompts instruct the LLM to analyze structured information and generate scores for various attributes, along with the rationale and tips on how to improve it by providing it with the name and description of each factor. All prompts are listed in the supplementary material.

2.1.1 Actionable Resume Feedback. ResumeGenAI evaluates the resume on the following attributes: overall impact, professional experience, skills and achievements, ATS compatibility, language and grammar, and visual appearance (Figure 1). Each of these attributes was chosen after carefully reviewing various online resume evaluation and scoring tools such as Jobscan [11], ResumeWorded [37] and others ([2], [39], [32], [12], [17], [5], [33]) and previous literature such as [14], [18], [36]. For each of these attributes, ResumeGenAI generates a score on a scale of 1 to 10. It offers insights into the score of each factor, detailing the reasoning behind it and providing suggestions for improving that score. (Figure 2).

2.1.2 Resume Personalization. Using this feature, job seekers can personalize their resume for a specific job role and job posting. ResumeGenAI provides additional feedback on job relevance, helping them understand how well their resume aligns with the target job, along with actionable suggestions for improvement.

2.1.3 Resume Comparison. This feature enables job seekers to compare two different resumes by visualizing a bar graph (Figure 3) of score comparisons in various evaluation attributes, accompanied by a summary of qualitative differences (Figure 4). It supports self-assessment by allowing users to track resume improvements over time and benchmark against resumes that align with the ideal requirements of a targeted job role.

2.1.4 Personalized Chat Assistant. This feature offers a conversational agent that provides real-time responses to job seekers’ queries (Figure 5a). It can be initiated with or without a saved resume analysis. When launched with a saved resume, the agent is initiated with resume details and feedback for personalized support; otherwise, it handles general resume-related questions. This helps users quickly resolve doubts, understand feedback, and receive guidance throughout the resume-building process.

2.1.5 Comprehensive Resource Center. ResumeGenAI provides job seekers with curated educational resources on effective resume building, including articles, best practices, and expert tips from HR professionals (Figure 5c and Figure 5d). It also provides professional Overleaf resume templates (Figure 5b) for effective resume formatting.

Additionally, ResumeGenAI allows job seekers to save their resume analysis on the platform or export it as PDF for sharing.

3 Evaluating ResumeGenAI with Job Seekers

To understand how job seekers perceive AI-generated feedback compared to traditional feedback on resumes, and to analyze whether it can enhance the quality of resumes, we conducted an exploratory study that included control and treatment groups. The study consisted of three phases: pre-treatment, three weeks of treatment, and post-treatment. Outcome variables were measured pre and post-treatment. These outcomes included expert review of resumes and participant feedback on their experience with the platform.

3.1 Participant recruitment and characteristics

We recruited the participants from a pool of university students. We reached out to all students via email asking for their voluntary participation in this study. We used the following screening criteria: (i) 18 years and older; (ii) actively seeking job opportunities; (iii) preparing for interviews. This criterion provided us with a pool of around 800 participants. 80 participants in this group showed interest in our study and completed our initial demographic form. Of the 80 participants who responded, 32 did not register on the platform, effectively making the number of participants 48.

Of the 48 participants, all were actively seeking a job, comprising 38 males, and 10 females, with a mean age of 21.95 years (SD = 3.01). After 3 weeks, 34 participants responded to the post-treatment session, representing a 29.16% attrition rate. 17 participants were retained in each control and treatment group. Among the 34 participants, 28 (82.35%) were male, and 6 were (17.65%) female, with a

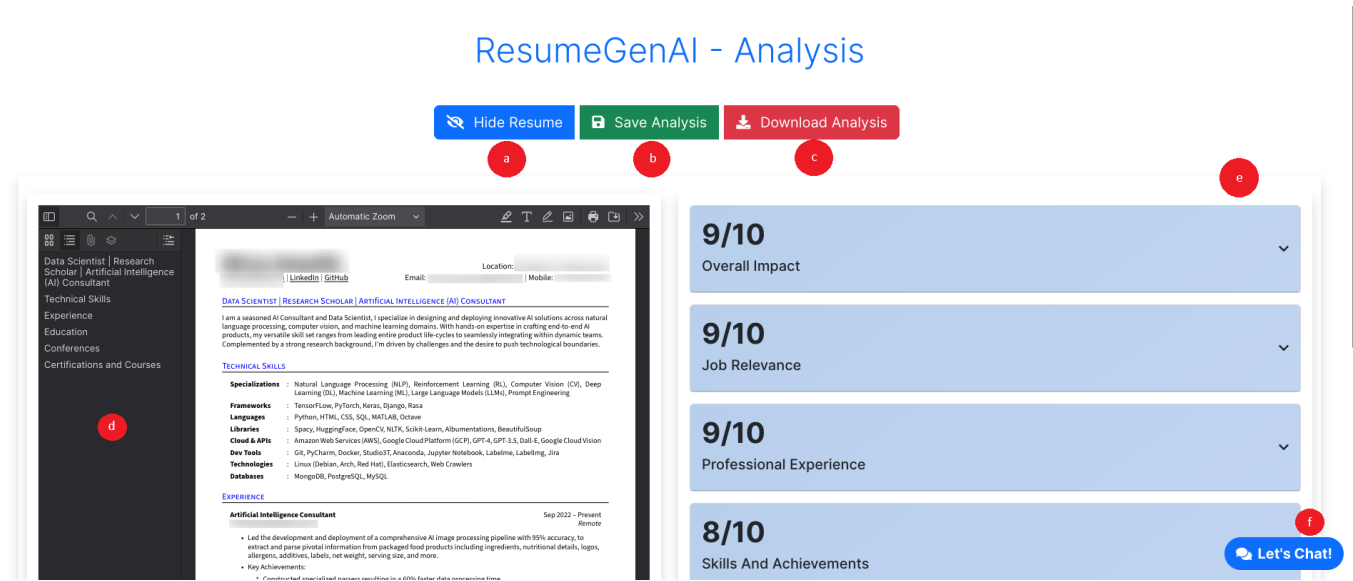


Figure 1: Resume analysis page on ResumeGenAI: (a) Hide/View resume on the left side. (b) Save resume analysis on the platform to view later. (c) Download analysis in a PDF format for offline use. (d) View of uploaded resume. (e) Analysis of resume; you can click on each factor to view detailed analysis with insights, and improvement tips. (f) Conversational agent personalized with resume.

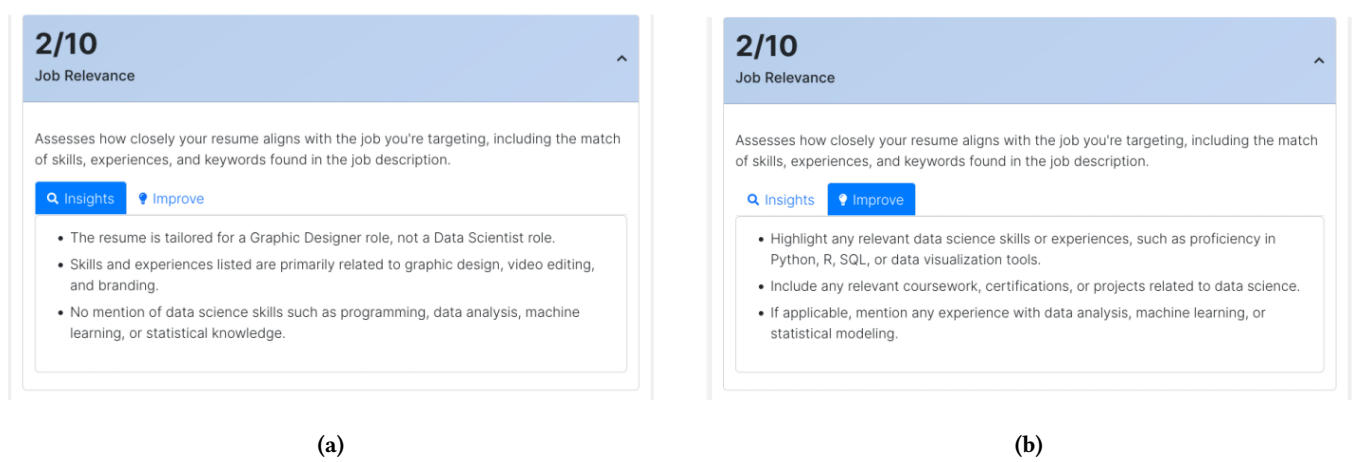


Figure 2: Detailed feedback tab of Job Relevance: (a) Detailed insights. (b) Actionable feedback on how to improve.

mean age of 22.2 years ($SD = 3.33$). In terms of educational background, 41.17% ($n = 14$) were graduates, 29.41% ($n = 10$) were post-graduates, 26.47% ($n = 9$) had completed their schooling, and 2.94% ($n = 1$) had a doctoral degree. 88.23% ($n = 30$) were freshers with no prior work experience, 5.55% ($n = 2$) had 1-3 years of experience, and 5.55% ($n = 2$) had more than 3 years of work experience. Among the participants, 61.76% ($n = 21$) reported having received feedback on their resume from an expert before, 35.29% ($n = 12$) had not received any feedback and 2.94% ($n = 1$) responded with "maybe." On the scale of 1-5 for rating their resume building skills, 55.88%

($n = 19$) rated 3, 23.52% ($n = 8$) rated 4, 17.64% ($n = 6$) rated 2, and 2.94% ($n = 1$) rated 1.

3.2 Controlled ResumeGenAI Version

The controlled version retains ResumeGenAI's intuitive web UI and shared resources, but delivers guidance in the form of carefully curated best practices based on literature [35], [26], [34], [21], [28], [9] rather than LLM-generated analysis.

- **Curated Feedback Lists.** A concise set of do's and don'ts for each core attribute — visual layout, skills presentation,

Compare Resumes

Prerequisite:

1. Upload your resume at [Analyse Resume](#) for analysis.

2. Save the analysis, later viewable at [Saved Analysis](#).

3. Compare any two saved resume analysis here.

Resume 1

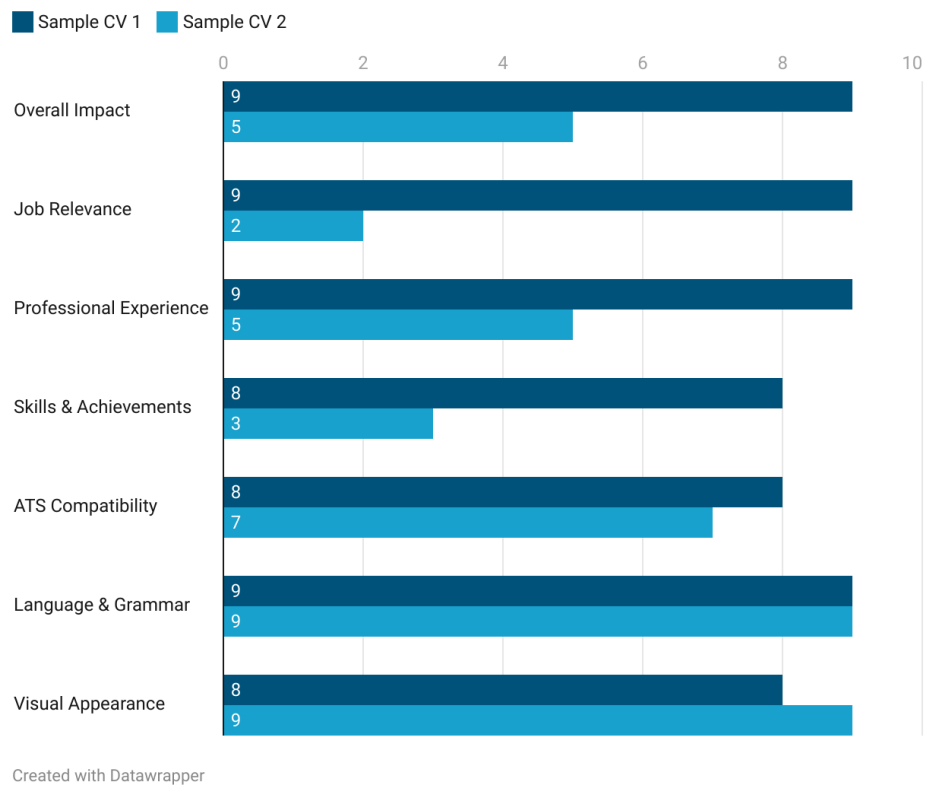
Sample CV

Resume 2

Sample CV II

Compare

(a)



(b)

Figure 3: Resume comparison feature: (a) Select resumes to compare. (b) Bar plot depicting difference in scores of two resumes.

experience description, grammar, and more — compiled from leading HR articles and tutorials.

- **Manual Comparison Guide.** Step-by-step detailed guide that explains how to compare two resumes and identify differences in structure, skills, content depth, relevance, etc.
- **Personalization Checklist.** A role-tailoring guide that outlines how to align your resume with a target job for example, selecting keywords, emphasizing relevant projects, and organizing sections.

- **Resource Center and Templates.** Access to the same curated articles, videos, and Overleaf resume templates available in the experimental version under comprehensive resource center.
- **Save & Export.** Ability to save your progress on the platform and export all guidance as a shareable PDF report.

Comparison Feedback

- Resume 1 is better than Resume 2 because it is highly relevant to the Data Scientist role, with extensive experience in AI, NLP, and machine learning, whereas Resume 2 is tailored for a Graphic Designer role.
- Resume 1 has a higher overall impact score (9) compared to Resume 2 (5), indicating stronger alignment with the desired job role and better documentation of relevant skills and achievements.
- Resume 1's professional experience is diverse and well-quantified with achievements relevant to data science, while Resume 2 lacks data-related projects or roles.
- Resume 1 includes certifications from reputable institutions that enhance credibility in the data science field, whereas Resume 2 does not mention any relevant certifications.
- Resume 1 is structured to be ATS-friendly with relevant keywords, while Resume 2 lacks crucial data science keywords, reducing its ATS compatibility for a Data Scientist role.
- Both resumes score equally high in language and grammar, but Resume 1's content is more relevant to the target role.
- Resume 1's skills and achievements are more aligned with data science, whereas Resume 2's skills are focused on graphic design, which is not relevant to the Data Scientist role.
- Although Resume 2 has a strong visual appearance, Resume 1 also maintains a clear and professional layout that supports its relevance and readability for the target role.

Figure 4: Detailed subjective feedback on comparison of two resumes.

3.3 Procedure

3.3.1 Pre-treatment session: The pre-treatment session included a questionnaire that collected demographic information, the skills of the participants in resume building, and if they had received feedback on their resume and how frequently they updated it. All participants ($N = 48$) also submitted their resumes at the pretreatment session.

3.3.2 Experimental groups: The participants were randomly assigned to one of the two groups, control and treatment, with 24 participants in each group. Participants were not informed about their designated groups.

3.3.3 Three-weeks deployment: After the pre-treatment session, the ResumeGenAI app was available online on a website for participants to use. All participants were informed that the app would be available for free use, as many times as needed, for a period of three weeks. During that period, they received timely reminders once a week that they could use the ResumeGenAI app to improve their resume. Participants in the treatment group used the full-featured version of ResumeGenAI app (Section 2.1), whereas those in the control group interacted with the controlled ResumeGenAI app version (Section 3.2). Control group users were also encouraged to explore and leverage additional tools and community resources beyond ResumeGenAI to further refine their resumes.

3.3.4 Post-treatment session: $N = 34$ participants responded to the post-treatment session, with 17 participants in each group. During the post-treatment session, participants completed a questionnaire

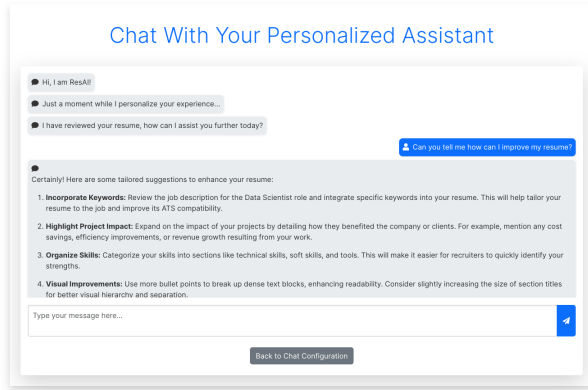
that evaluated each feature they had access to, as well as the System Usability Scale (SUS). We also collected their resumes, which were updated based on the feedback received from the app.

3.3.5 Expert rating procedure: We recruited five experts with experience in resume evaluations interviewing. The experts reviewed the resumes submitted by participants in both the pre-treatment and post-treatment sessions. Four experts reviewed the resumes of seven participants and one expert reviewed the resumes of six participants, giving a mean of 6.8 participants per expert. The experts were blinded to both the background of the participants and the session (pre-treatment or post-treatment) from which each resume originated, ensuring unbiased evaluations.

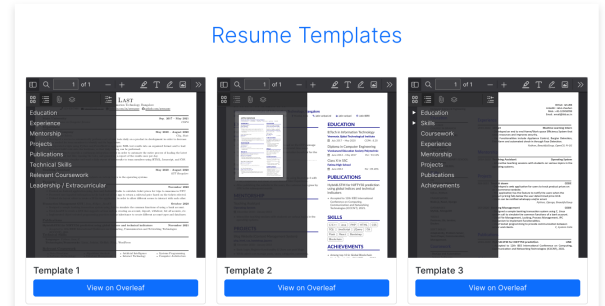
3.4 Metrics

3.4.1 Self-reported metrics. In the post-treatment session, participants in the control and treatment groups were asked to complete a questionnaire evaluating their overall experience with the platform and its features. The questionnaire also included items from the System Usability Scale (SUS) [1] to assess usability perceptions. All responses were recorded on a seven-point Likert scale. Table 1 summarizes the questions posed to both groups.

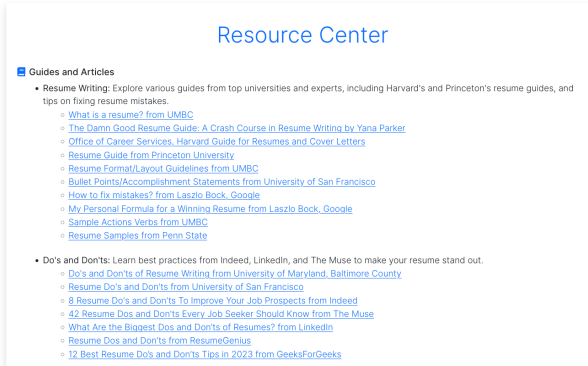
3.4.2 Expert rated metrics. In addition to self-reported metrics, experts rated each resume according to standardized criteria derived from an extensive review of research articles, widely used resume evaluation tools such as Enhancv [5], Resume Worded [37], Zety [39], etc. and prominent skill databases such as e.g., ESCO [4], O*NET [6], etc. Consultations with experienced HR professionals



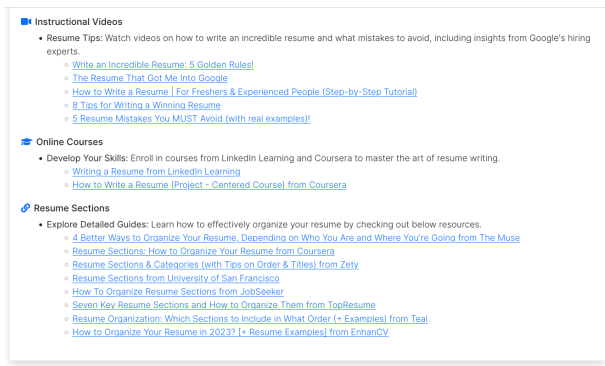
(a) Personalized chat assistant



(b) Resume templates



(c) Resource center I



(d) Resource center II

Figure 5: A Glimpse at Some of ResumeGenAI's Features.

ID	Question	P-Value	Z-Statistic
Q1	Overall Platform Experience	0.108	-1.537
Q2	Resume Comparison Feature	0.075	-1.707
Q3	Conversation Agent Feature	—	—
Q4	Personalized Resume Feature	0.004**	-2.708
Q5	PDF Export Feature	0.001***	-2.781
Q6	Ease of Implementing Suggestions	0.209	-1.244
Q7	Resume Templates	0.028*	-2.098
Q8	Resource Center Quality	0.232	-1.146
Q9	Likelihood of Recommendation	0.408	-0.756
Q10	Trust in Provided Feedback	0.336	-0.927

Table 1: Summary of post-treatment questionnaire responses comparing control and treatment groups. The *P-Value* column indicates the statistical significance of differences in responses between the two groups for each item. Note: * $p < 0.05$, ** $p < 0.01$, * $p < 0.001$**

further refined the final set of metrics. Each metric was measured on a seven-point Likert scale, covering dimensions such as completeness, clarity, and overall impact. The metrics rated by experts are summarized in Table 2.

4 Results

This section presents the main findings of our evaluation, drawing from both self-reported metrics from study participants and domain-expert analyzes of resumes. We begin by detailing participant feedback, including participant ratings of various platform

ID	Metric Name	Metric Definition	P-Value	Z-Statistic
M1	Completeness	The extent to which the resume includes all necessary sections and information.	0.100	1.532
M2	Skills	The extent to which the resume highlights the skills of the candidate.	0.025*	2.083
M3	Achievements	The extent to which the resume highlights the achievements of the candidate.	0.295	1.016
M4	Content Quality	The quality and depth of the content provided in the resume.	0.201	1.222
M5	Clarity	The ease with which the content can be understood by the reader.	0.144	1.412
M6	Language	The degree to which the resume reflects a professional language throughout the resume.	0.162	1.326
M7	Grammar	The degree to which the resume is grammatically correct throughout the resume.	0.023*	2.135
M8	Consistency	The uniformity of formatting throughout the resume.	0.151	1.377
M9	Visual Appearance	The overall aesthetic of the resume, including layout, spacing, and use of design elements.	0.292	1.033
M10	Overall Impact	The extent to which the resume highlights the skills of the candidate.	0.020*	2.152

Table 2: Expert-rated resume evaluation metrics (7-point Likert scale). P-values are from Mann-Whitney U tests comparing post-intervention scores between the treatment ($n = 17$) and control ($n = 17$) groups. Controlled (pre-intervention) comparisons were non-significant for all metrics. Note: * $p < 0.05$, ** $p < 0.01$, * $p < 0.001$**

features and overall usability perceptions. We then turn to expert evaluations, in which industry professionals rated the quality of resumes based on a comprehensive set of criteria defined in Table 2. Through these complementary data sources, we assess the impact of AI-driven interventions on user experience, resume quality, and the potential benefits of integrating such automated guidance into the resume writing process.

4.1 Self-Reported Metrics

4.1.1 Participant Experience Feedback. Table 1 summarizes the evaluations of the participants on various characteristics of the platform reported after completing the study. The control and treatment groups each provided self-reported ratings on elements such as overall platform experience, resume comparison, personalized resume creation, PDF export functionality, and the likelihood of recommending the system to others.

To assess whether there were systematic differences in these ratings between the two groups, a Mann-Whitney U Test [19] was performed. Because the data did not meet the assumptions for normality, this nonparametric test was deemed appropriate. The results, mentioned in Table 1, indicate statistically significant differences for three characteristics: (i) *Resume Personalization*; (ii) *Save & Download Analysis*; and (iii) *Resume Templates*. Additionally, the *Resume Comparison* feature approached, but did not reach, statistical significance. These findings suggest that participants in the treatment group found certain features - particularly those that facilitated customization and simplified document handling - to be more impactful or beneficial compared to the control group.

4.1.2 System Usability Scale (SUS) Evaluation. The System Usability Scale (SUS) is a widely used to measure perceived usability,

giving scores on a scale of 0-100. A benchmark value for SUS is typically reported as 68, which serves as a point of reference for assessing overall usability.

In our study, the treatment group achieved an average SUS score of 80.42, significantly higher than the conventional benchmark. The control group reported a mean score of 72.92, also exceeding the standard average, although to a lesser extent. When aggregated, both groups produced a combined SUS score of 76.67, positioning the system firmly above the commonly referenced usability threshold of 68. These findings suggest that participants generally perceived the application as user-friendly, with clear navigation, intuitive functionality, and minimal barriers to task completion.

The comparatively higher usability rating in the treatment group indicates that the interventions introduced in the treatment condition may have positively influenced user perceptions of efficiency and ease of use. Although the control group score still exceeded the average, the evident gap between the two groups highlights the potential effectiveness of AI-generated feedback in the treatment condition.

4.2 Expert-Rated Resume Evaluations

Table 2 outlines ten different metrics used by domain experts to evaluate the resumes of participants, each rated on a seven-point Likert scale.

A Mann-Whitney U test was performed on the pre and post-intervention resume scores for both the control ($n = 17$) and treatment ($n = 17$) groups. No significant group differences were observed in the pre-intervention. However, the analysis revealed statistically significant improvements in post-intervention scores in three key dimensions for participants in the treatment group: (i)

Skills (M2); (ii) *Grammar (M7)*; and (iii) *Overall Impact (M10)* (see Table 2).

These findings suggest that the approach of the treatment group to resume writing more effectively highlighted their relevant skills, demonstrated stronger grammatical precision, and ultimately presented a more compelling overall narrative. Although other metrics (e.g., completeness, clarity, consistency) did not achieve statistical significance, the upward trend in these measures within the treatment group underscores the broader positive influence of the intervention.

In particular, the improvement in the Skills (M2) indicates that the participants may have been more systematic in identifying and articulating their competencies after utilizing ResumeGenAI. Likewise, the improvements in 'Grammar' (M7) point to increased attention to detail and correctness, an important factor for resumes that aim to create a strong first impression. Finally, the gains in 'Overall Impact' (M10) signal a more cohesive presentation, encompassing not only the content and layout of the resume, but also how effectively it conveys the professional story of the candidate.

5 Discussion

Job seekers can use *ResumeGenAI* to boost their chances of getting through initial resume screening and ATS applications. Whereas job seekers once had to wait for peers or experts to review their resumes, they can now get instant feedback. They can get insights into their resume in real-time, and personalized detailed feedback on how to improve the resume. They can reiterate this process as many times as they want, until they get a satisfactory score for their resume. If they are unsure of which resume templates to use, ResumeGenAI provides them with the templates. If they need more help with the feedback, ResumeGenAI has an integrated conversational agent that can clarify any questions, for example, rephrasing content, optimising keywords, quantifying achievements with metrics and restructuring for clarity. It is also evident from our study that the resumes generated by participants of the treatment group received a higher overall impact, skills, and grammar scores compared to the control group, as measured by the experts. This means the treatment group participants had better resumes, which presented their skills well, were grammatically more correct and structured, and had a better overall influence on the recruiters.

While this technology empowers job seekers, it can also lead to homogenous resumes or encourage overstatement [3]. Users should leverage AI responsibly—personalizing their narrative and verifying every claim to ensure authenticity throughout the hiring process. Over-reliance on AI feedback can also downgrade one's own writing and personal-branding skills, introducing algorithmic biases that may undervalue certain occupations or underrepresented individuals from specific ethnic or demographic backgrounds [40]. It can also pose privacy risks to users as personal identifiable information (PII) is submitted to third-party platforms.

For employers, AI-augmented resumes offer advantages in streamlining candidate screening - standardized, keyword-rich documents integrate seamlessly with applicant-tracking systems and can markedly reduce time-to-hire. Yet, this very standardization may mask indicators of creativity or unique skill sets [8], while

hyper-optimized submissions carry the risk of exaggerated credentials. As a result, recruiters can increasingly offset these limitations by emphasizing in-depth interviews, practical assessments, or work sample tests, potentially extending later stages of the hiring cycle. Moreover, the processing of sensitive applicant Personally Identifiable Information (PII) through third-party AI platforms raises significant privacy and ethical concerns, mandating robust data-governance frameworks and strict adherence to regulatory requirements. Finally, if advanced AI tools remain accessible only to a privileged subset of applicants, technological disparities could deepen existing inequities in the talent pool [30], highlighting the imperative to promote broad, equitable access and support for all candidates.

6 Conclusion

In this work, we introduced *ResumeGenAI*, a platform that leverages LLM capabilities to provide actionable feedback on resumes and improve the user experience through additional features. Our user study indicates that AI-generated feedback positively influences resume quality of job seekers, as assessed by domain experts. Future work will focus on extending the capabilities of ResumeGenAI to automatically edit resumes based on feedback, further automating and streamlining the resume writing process.

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